Cluster Algebras

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How strange to actually have to see the path of your journey in order to make it.

—Neal Shusterman, [Shu16]

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MINICOURSE 1

CLUSTER ALGEBRAS AND CLUSTER VARIETIES

Did I mention that at least 50% of the signs are wrong? Because I am doing worse than chance.

—Sophie Morel

These talks were given by Robin Zhang.

1.1 April 22

We only have three lectures, so we will have to be fast. We will try to introduce the basic notions, and then we will try to explain their applications to Diophantine equations. Here is a rough plan.

1. We'll start with some history and introduce some combinatorics which give rise to cluster algebras $\mathcal{A}$. The combinatorics arises by studying maps $\mathcal{A} \to \mathbb{Z}$.
2. Then we will introduce the basic notions of cluster algebras.
3. We will explain the classification of cluster algebras.
4. With cluster algebras in hand, we will define cluster varieties, which look like $\operatorname{Spec} \mathcal{A}$.
5. From here, we will be able to discuss applications to positive integral points on the affine varieties $\operatorname{Spec} \mathcal{A}$.
6. Lastly, we will explain applications to arithmetic dynamics (in "Donaldson–Thomas dynamics") on the cluster varieties.

1.1.1 Classical Friezes

Cluster algebras began their life in the theory of total positivity and representations of quivers. They later found applications in Teichmüller theory and symplectic/Poisson geometry.[1] They can also be found in Grassmannians, in quantum field theory, and cosmology. We will say nothing about any of these applications. Instead, we will explain connections to counting (positive) integral points on log Calabi–Yau varieties and arithmetic dynamics, originally discovered through the Markov surface.

Cluster algebras arise from freizes. Here is the definition of a classical frieze.

[1] Pavel Etingof points out that the cluster algebras found in Poisson geometry are algorithmically quantizable, which provide an interesting source of examples of quantization.

Definition 1.1 (frieze)**.** A *frieze* is the central section of an entablature above a Greek temple.

Example 1.2. The Parthenon featured a frieze depicting historical/mythological scenes.

This definition is slightly too classical. Nonetheless, it will be useful to keep in mind that an architectural frieze usually featured a horizontally periodic geometric pattern.

Definition 1.3 (frieze)**.** A *frieze pattern* is an array of positive integers with infinitely many columns and some SL_2-unimodular conditions. More explicitly, a frieze pattern with n rows is an array $\{x_{ij}\}_{ij}$ where $i \in \{1, 2, \ldots, n\}$ and $j \in \mathbb{Z}^+$ satisfying the condition

$$x_{ij}x_{i,j+1} - x_{i-1,j+1}x_{i+1,j} = 1.$$

Remark 1.4. One typically arranges the frieze as follows.

$$\begin{array}{ccccccccccccc}
\cdots & 1 & & 1 & & 1 & & 1 & & 1 \cdots & & \\
\cdots & & x_{11} & & x_{12} & & x_{13} & & x_{14} & & \cdots & \\
\cdots & x_{2,-1} & & x_{21} & & x_{22} & & x_{23} & & x_{24} & \cdots & \\
\cdots & & \ddots & & \ddots & & \ddots & & \ddots & & \cdots & \\
\cdots & 1 & & 1 & & 1 & & 1 & & 1 & & 1 & \cdots
\end{array}$$

With this convention, we see that $x_{ij}x_{i,j+1} - x_{i-1,j+1}x_{i+1,j} = 1$ is asking for each "diamond" to have determinant 1.

Example 1.5. Here is a Frieze pattern with 2 rows.

$$\begin{array}{ccccccccccccc}
\cdots & 1 & & 1 & & 1 & & 1 & & 1 & & 1 & & \cdots \\
\cdots & & 3 & & 1 & & 2 & & 2 & & 1 & & 3 & \cdots \\
\cdots & 1 & & 2 & & 1 & & 3 & & 1 & & 2 & & \cdots \\
\cdots & & 1 & & 1 & & 1 & & 1 & & 1 & & 1 & & \cdots
\end{array}$$

(We do not count the row of ones in the number of rows.) We note that there is a "fundamental domain" with $(3, 1, 2)$ in the first row and $(2, 1)$ in the bottom row.

Gauss (roughly speaking) proved the following classification.

Theorem 1.6 (Gauss 1866)**.** All friezes with two rows are 5-periodic.

Remark 1.7. This was proven "posthumously" as Gauss's theorem on the pentagramma minificum; Gauss proved some purely geometric fact which turns out to imply that any three of the equations

$$\begin{cases} 1 + \alpha = \gamma\delta, \\ 1 + \beta = \delta\varepsilon, \\ 1 + \gamma = \alpha\varepsilon, \\ 1 + \delta = \alpha\beta, \\ 1 + \varepsilon = \beta\gamma \end{cases}$$

implies the other two, which Coxeter recognized classifies friezes.

Coxeter would go on to try to classify these friezes.

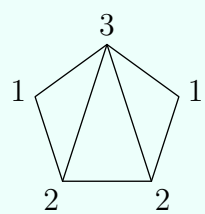
Theorem 1.8 (Coxeter 1971)**.** All friezes with n rows are periodic with period $n+3$.

Remark 1.9. Periodicity continues to hold, even if we do not require that the entries of the frieze are integers.

Theorem 1.10 (Conway–Coxeter 1973)**.** Fix n. Then there are exactly $\frac{1}{n+2}\binom{2n+2}{n+1}$ friezes with n rows.

Sketch. Note that $\frac{1}{n+2}\binom{2n+2}{n+1}$ is a Catalan number, so in Catalan tradition, we will try to find a bijection with one of the many other ways to count Catalan numbers. It turns out that we can friezes with n rows can be found in bijection with triangulations of $(n+3)$-gons. In particular, given a triangulation of $(n+3)$-gons with labeled vertices $\{1, 2, \ldots, n+3\}$, define $\{x_{1j}\}_j$ to be the periodic sequence of the number of triangles which hit vertex j. Then one can use the determinant condition to complete the frieze. Injectivity of this map follows because one can recover the triangulation from the sequence, and this turns out to hit all friezes, which one can check by induction on n. ■

Example 1.11. For $n = 2$, there is only one triangulation of the pentagon (up to rotation), so we find that Example 1.5 gives the only frieze up to translation.



Example 1.12. There is a triangulation of the nonagon given by the sequence $(4, 2, 1, 4, 2, 1, 4, 2, 1)$. This provides an example of a frieze with six rows with this sequence as the first row.

1.1.2 General Friezes

Here are some more general friezes; this definition is due to Caldero and Chapoton.

Definition 1.13 (frieze)**.** Fix a Dynkin type Δ with (generalized) Cartan matrix $C = \{c_{ij}\}_{i,j\leq n}$. Then a *frieze of type* Δ is an array $\{x_{ij}\}$ with $i \in \{1, 2, \ldots, n\}$ and $j \in \mathbb{N}$, satisfying the condition

$$x_{ij}x_{i,j+1} - \prod_{k<i} x_{k,j+1}^{|c_{ki}|} \prod_{k>i} x_{kj}^{|c_{ki}|} = 1.$$

Example 1.14. Taking $\Delta = A_n$ recovers friezes with n rows.

Example 1.15. Type E_8 has the following Dynkin diagram.

This means that the frieze should have seven rows, but the third row will have extra numbers in it ("filling in the diamond"), which must then satisfy the condition $ae - bcd$, where these numbers fit into the following diamond.

$$\begin{array}{ccc} & b & \\ a & c & e \\ & d & \end{array}$$

There is a frieze with

$$(x_{11}, \ldots, x_{18}) = (4, 11, 29, 5, 21, 13, 5, 2),$$

where the vertices are colored as above.

We now try to prove the same theorems as Coxeter (and Conway).

Theorem 1.16 (Assem–Reutenauer–Smith)**.** Fix a Dynkin type Δ. Then all friezes of type Δ are periodic if and only if Δ is of finite type.

Remark 1.17. The periods are known for all Dynkin types. For example, it is 16 for type E_8.

With some periodicity, we can try to count these things. We have little to say about infinite types.

Theorem 1.18. If Δ is a Dynkin type of infinite type, then there are infinitely many friezes of type Δ.

Remark 1.19. It turns out that the coefficients of a frieze are unbounded in infinite types.

For finite types, we can count them.

Theorem 1.20. Fix a Dynkin type Δ of finite type. Then the friezes of type Δ can be enumerated as follows.

Δ	A_n	B_n	C_n	D_n	E_6	E_7	E_8	F_4	G_2
$\#$	$\frac{1}{n+2}\binom{2n+2}{n+1}$	$\sum_{m=1}^{\lfloor\sqrt{n+1}\rfloor}\binom{2n-m^2+1}{n}$	$\binom{2n}{n}$	$\sum_{m=1}^{n} d(m)\binom{2n-m-1}{n-m}$	868	4400	26952	112	9

Remark 1.21. The counts in type E_7 and E_8 were conjectured in 2016 (by computer search) but only proven recently by Robin Zhang.

Remark 1.22. We will say basically nothing about the proof (for now). Many of the arguments find bijections between these friezes and some kind of polygon triangularizations, perhaps with punctures and so on. The arguments for E_7 ad E_8 are more arithmetic in nature, and they will be discussed later.

Remark 1.23. The enumerations for E_7 and E_8 are due to Robin Zhang. He used cluster algebras (and Diophantine aspects of cluster varieties).

Of course, we have yet to define a cluster algebra, but we can give some motivation for them via the following counting.

Proposition 1.24. Fix a Dynkin type Δ, and let $\mathcal{A}$ be a cluster algebra of type Δ. Then the friezes of type Δ are in bijection with ring homomorphisms $f\colon \mathcal{A} \to \mathbb{Z}$ which map the cluster variables in $\mathcal{A}$ to positive integers.

1.1.3 Cluster Algebras

We are now allowed to define cluster algebras. The most general definition is technical for the moment, so we will content ourselves with the following notion, postponing a formal definition until later.

Almost Definition 1.25 (cluster algebra)**.** A *cluster algebra* $\mathcal{A}$ over a field k is a commutative subring of $\mathbb{Q}(x_1, \ldots, x_n)$ equipped with *cluster variables* which are constructed from some mutation $n \times n$ matrix B as follows (with zeroes along the diagonals).

1. Start with $(x_1, \ldots, x_n)$. Then define

$$x_1' := \frac{\prod_{\substack{1 \le i \le n \\ b_{i1} < 0}} x_i^{|b_{i1}|} + \prod_{\substack{1 \le i \le n \\ b_{i1} > 0}} x_i^{|b_{i1}|}}{x_1}.$$

   In words, in the first column, we take a weighted product of the x_is with positive coefficients, and the weighted product of the x_is with negative coefficients. Lastly, we divide by x_1. We then change B by flipping the sign on the first row and first column. This gives a new tuple of rational functions $(x_1', x_2, \ldots, x_n)$.

2. We then repeat the process at x_2, then x_3, and so on. Eventually we go back and repeat.

At the end, $\mathcal{A}$ is generated by the resulting rational functions.

Example 1.26. Over the field $\mathbb{Q}$, we define a cluster algebra $\mathcal{A} \subseteq \mathbb{Q}(x_1, x_2)$ with the mutation matrix $B := \left[\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right]$. Then $\mathcal{A}$ is defined as follows.

1. We take the pair (x_1, x_2) and apply the following process: take x_1, and define a numerator to be the product of all variables with a positive coefficient in B (in the first column) plus the product of the variables with a negative coefficient, and then we define out by x_1. This gives $x_1' = (x_2+1)/x_1$. Lastly, we adjust B by flipping signs in the second row and second column, giving $B = \left[\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right]$.

2. Then we repeat the process with the second column. This gives $x_2' = (x_1' + 1)/x_2 = (x_1 + x_2 + 1)/(x_1 x_2)$ and $B'' = \left[\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right]$.

3. We now go back to do the process to x_1, which gives $x_1'' = (x_2' + 1)/x_1' = (x_1 + 1)/x_2$.

4. Repeating the process for the second column gives $x_2'' = (x_1'' + 1)/x_2' = x_1$.

5. Going back to the first column gives $x_1''' = x_2$.

Thus, we see that our process is periodic! The resulting cluster algebra is thus generated by five rational functions (which are the cluster variables), each of which is a cluster variable.

Remark 1.27. One can play the same game with any skew-symmetric matrix B with integral coefficients. In fact, it is even enough if B is skew-symmetrizable, meaning that it is the product of a skew-symmetric matrix and a diagonal matrix with positive entries.

We will classify cluster algebras next time. The above example will turn out to be type A_2; the number of cluster variables turns out to correspond to the period of the frieze.

1.2 April 27

Today, we classify cluster algebras.

1.2.1 Cluster Algebras, Again

Last time, we constructed a cluster algebra attached to a skew-symmetrizable matrix. Let's return to this definition, as follows.

Definition 1.28 (mutation)**.** Fix a positive integer n. A *seed* is a pair (x, B) of an n-tuple $x = (x_1, \ldots, x_n)$ of algebraically independent variables over $\mathbb{Q}$, and a *mutation matrix* B is an $n \times n$ skew-symmetrizable matrix. Given this data, we define the kth *mutation* $\mu_k(x, B)$ to be the seed given by

$$\mu_k(x)_i := \begin{cases} x_i & \text{if } i = k, \\ \dfrac{\prod_{b_{jk}>0} x_j^{|b_{jk}|} + \prod_{b_{jk}<0} x_j^{|b_{jk}|}}{x_k} & \text{if } i \neq k, \end{cases}$$

together with the mutation matrix given by the matrix B' with coefficients

$$B'_{ij} = \begin{cases} -b_{ij} & \text{if } i = k \text{ or } j = k, \\ b_{ij} + \frac{|b_{ik}|b_{kj} + b_{ik}|b_{kj}|}{2} & \text{if } i \neq k \text{ and } j \neq k. \end{cases}$$

Remark 1.29. By construction, applying the kth mutation twice does nothing. This property explains the rule for mutating the mutation matrix. Roughly speaking, the mysterious second equation in the mutation of B should be split up into two cases: if b_{ik} and b_{kj} have the same sign, then the two pieces combine; and if they have different signs, then we have cancellation.

Definition 1.30 (cluster algebra)**.** Fix a positive integer n, and choose a seed (x, B). Then the associated *cluster algebra* $\mathcal{A}$ is the $\mathbb{Z}$-subalgebra of $\mathbb{Q}(x)$, generated by all possible sequence of mutations. The collection of generators is known as the set of *cluster variables*.

Remark 1.31. Our cluster algebras $\mathcal{A}$ will have $\mathcal{A} \cap \mathbb{Q} = \mathbb{Z}$, which follows from Theorem 1.37.

It is not at all obvious if the associated cluster algebra is finitely generated. In light of our cluster variables, the correct notion is as follows.

Definition 1.32 (finite type)**.** A cluster algebra $\mathcal{A}$ has *finite type* if and only if there are only finitely many distinct cluster variables.

Here is how we will test this.

Definition 1.33. A cluster algebra $\mathcal{A}$ is *acyclic* if and only if the associated seed (x, B) (and its mutations) produce directed graphs with no oriented cycles. Here, the directed graph associated to an $n \times n$ skew-symmetric B is the one with vertices $\{v_1, \ldots, v_n\}$ and b_{ij} edges from v_i to v_j; for a skew-symmetrizable matrix DB where D is diagonal, then we also label the edges by the diagonal elements: the edge $i \to j$ for $b_{ij} > 0$ is labeled by the pair $(|b_{ij}|, |b_{ji}|)$ when $b_{ij} > 0$.

Remark 1.34. It turns out that $\mathcal{A}$ is finitely generated if $\mathcal{A}$ is acyclic.

Example 1.35. The matrix $\left[\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right]$ induces the graph $1 \to 2$. The only mutation of this matrix is its negative, so we see that this cluster algebra is acyclic.

Non-Example 1.36. The matrix

$$\begin{bmatrix} 0 & 2 & -2 \\ -2 & 0 & 2 \\ 2 & -2 & 0 \end{bmatrix}$$

produces the graph

$$\begin{tikzcd} 1 \arrow[r, shift left] & 2 \arrow[l, shift left] \arrow[d, shift left] \\ & 3 \arrow[ul, shift left] \arrow[u, shift left] \end{tikzcd}$$

which is notably not acyclic. The mutations of this matrix correspond to Vieta jumps. Namely, specializing the seed variables $(x_1, x_2, x_3) \mapsto (1, 1, 1)$ discovers all solutions to the Markov equation $x^2+y^2+z^2 = 3xyz$.

1.2.2 Theorems on Cluster Algebras

Here are the main theorems on cluster algebras. The following is due to Fomin and Zelevinsky.

Theorem 1.37 (Laurent phenomenon)**.** Fix a cluster algebra $\mathcal{A}$ with initial seed $((x_1, \ldots, x_n), B)$. Then any cluster variable in $\mathcal{A}$ is a Laurent polynomial in the variables $(x_1, \ldots, x_n)$. In other words, $\mathcal{A} \subseteq \mathbb{Z}[x_1, \ldots, x_n, 1/x_1, \ldots, 1/x_n]$.

Sketch. The key ingredient is the following intersection lemma: if x mutates to x' via the ith mutation, then by construction, $x_i x_i'$ can be expanded into a sum of two disjoint monomials. Thus, setting $R := \mathbb{Z}[x_1^{\pm}, \ldots, x_{i-1}^{\pm}, x_{i+1}^{\pm}, \ldots, x_n^{\pm}]$, we find

$$R\left[x_i^{\pm}\right] \cap R\left[(x_i')^{\pm}\right] = R[x_i, x_i'].$$

For example, the inclusion from the right to the left follows from the expansion of $x_i x_i'$. The moral is that having a cluster variable z which is a Laurent polynomial in the variables $(x_1, \ldots, x_n)$ and also a Laurent polynomial in all adjacent mutated seeds, then $z \in \mathbb{Z}[x_1, \ldots, x_n, x_1', \ldots, x_n']$, so z is Laurent.

One can then do an induction in the "distance" of a cluster variable from the initial seed. Namely, at distance zero, there is nothing to do. At distance one, this follows from the construction. At distance two, one has a cluster variable which is Laurent in distance one, and one can check by hand that the resulting polynomials are also Laurent in all the other mutations in distance two (and hence polynomial), which again uses the given mutation formula explicitly. After this, the induction kicks in: for example, in distance three, one can use the distance-two version to show that a given cluster variable is a polynomial in cluster variables that are distance zero and one. ■

Remark 1.38. Fomin called this proof "elementary but technical but elementary."

Remark 1.39. Here is an application: if we set $x_1 = \cdots = x_n = 1$, then all the cluster variables will be sent to integers. This remains true as long as the x_is are sent to elements of a ring which are units.

The following is due to many people, starting with Lee–Schiffler and ending with Gross–Hacking–Keel–Kontesevich. It uses scattering diagrams from mirror symmetry, which we do not have time for, so we will say nothing about its proof.

Theorem 1.40 (Positivity phenomenon)**.** Fix a cluster algebra $\mathcal{A}$ with initial seed $((x_1, \ldots, x_n), B)$. Then any cluster variable in $\mathcal{A}$ is a Laurent polynomial in the variables $(x_1, \ldots, x_n)$ with nonnegative coefficients.

Remark 1.41. The recursion can directly show that these polynomials take positive values on positive points.

Our last result is classification.

Definition 1.42 (Cartan companion)**.** Fix a skew-symmetric matrix B. Then we define the *Cartan companion* C to be the matrix C with coefficients

$$c_{ij} := \begin{cases} 2 & \text{if } i = j, \\ -|b_{ij}| & \text{if } i \neq j. \end{cases}$$

Example 1.43. The Cartan companion of the matrix $\left[\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right]$ is the matrix $\left[\begin{smallmatrix} 2 & -1 \\ -1 & 2 \end{smallmatrix}\right]$.

Theorem 1.44. Fix a cluster algebra $\mathcal{A}$ with initial seed $((x_1, \ldots, x_n), B)$. Then $\mathcal{A}$ has finite type if and only if B and all of its iterated mutations has positive-definite Cartan companion.

Sketch. In rank 2, our B looks like $\left[\begin{smallmatrix} 0 & b \\ -c & 0 \end{smallmatrix}\right]$, and we are interested in what happens to (x_1, x_2) when we alternatively apply the two mutations μ_1 and μ_2. This gives a sequence $\{x_1, x_2, x_3, \ldots\}$ of cluster variables. They satisfy

$$x_{m-1}x_{m+1} = \left[\begin{smallmatrix} x_m^b + 1 & \text{if } m \text{ is odd,} \\ x_m^c + 1 & \text{if } m \text{ is even.} \end{smallmatrix}\right]$$

One finds that the sequence $\{x_n\}$ is periodic (which is equivalent to finite type in this case because there are not so many mutations) if and only if $|bc| \leq 3$: the backward direction is a direct calculation, and the forward direction follows by some degree argument.

We now move on to higher rank. Let's start with the forward direction. We will show that if the Cartan companion avoids one of the finite Dynkin types fails to have finite type. Well, one should start by checking that B does not admit one of the rank-2 situations. It turns out that all affine Dynkin types (away from the finite type Dynkin types) and other infinite types have one of these infinite-type rank-2 subalgebras.

Lastly, we must show that B having finite Dynkin type implies that $\mathcal{A}$ has finite type, which is some direct calculation. Let Φ be the associated finite root system, and we choose a polarization with simple roots $\Pi = \{\alpha_1, \ldots, \alpha_n\}$. Now, for any cluster variable z, Theorem 1.37 tells us that

$$z = \frac{P(x_1, \ldots, x_n)}{x_1^{d_1} \cdots x_n^{d_n}},$$

where P is not divisible by any x_i with $d_i \neq 0$. We can now define a map which sends the cluster variables to $\operatorname{span} \Phi$ which sends $x_i \mapsto -\alpha_i$ for each i and in general sends z to $\sum_i d_i \alpha_i$. It turns out that z gets sent to a positive root by this map, and it turns out that this map is injective, so there are only finitely many cluster variables. In total, we find that the cluster variables are in bijection with $\Phi^+ \sqcup -\Pi$, so we being finite type follows. ■

Example 1.45. One can check that Example 1.26 is the cluster algebra from type A_3.

Non-Example 1.46. The seed $\left(x_1, x_2, \left[\begin{smallmatrix} 0 & -3 \\ 3 & 0 \end{smallmatrix}\right]\right)$ has all mutations of the mutation matrix with Cartan companion $\left[\begin{smallmatrix} 2 & 3 \\ 3 & 2 \end{smallmatrix}\right]$, which fails to be positive-definite. Indeed, this matrix admits a negative eigenvalue.

Remark 1.47. It turns out that the Cartan companions being positive-definite can be checked from the initial seed. This follows from the property of its mutation.

Remark 1.48. If $\mathcal{A}$ is acyclic, then $\mathcal{A}$ is finite type if and only if the initial seed's mutation matrix B has Cartan companion matrix of finite type. One needs to check the mutations only when $\mathcal{A}$ fails to be acyclic.

1.3 April 29

Here we go.

1.3.1 Cluster Varieties

For today, we will choose a finite type acyclic cluster algebra $\mathcal{A}$, for which we may assume has initial seed (x, B) with B whose lower-triangular submatrix is given by the associated Cartan matrix.

Definition 1.49 (cluster variety)**.** An *acyclic cluster variety* is an affine variety isomorphic to $\operatorname{Spec} \mathcal{A}$ for some acyclic cluster algebra $\mathcal{A}$. In the sequel, we will abbreviate this to a *cluster variety*.

Remark 1.50. There is a more general definition due to Fock–Goncharov and Gross–Hacking–Keel: namely, one can define a cluster variety to be covered by some algebraic tori T_Σ, which we glue together via some exchange relations. For example, there exist projective cluster varieties.

It will be helpful to have a version of a cluster variety which is manifestly integral and finite type.

Definition 1.51 (bounded cluster algebra)**.** Fix a cluster algebra $\mathcal{A}$ with initial seed (x, B). Then the *lower bound cluster algebra* $\mathcal{L}$ of $\mathcal{A}$ is the subalgebra generated over $\mathbb{Z}[x_1, \ldots, x_n]$ by the cluster variables which are one mutation away. Similarly, we define the *upper bound cluster algebra* $\mathcal{U}$ of $\mathcal{A}$ is the intersection

$$\mathcal{U} := \bigcap_z \mathbb{Z}\left[z_1^{\pm}, \ldots, z_n^{\pm}\right],$$

where z varies over all seeds.

Remark 1.52. Note that $\mathcal{L} \subseteq \mathcal{A} \subseteq \mathcal{U}$. Indeed, $\mathcal{L} \subseteq \mathcal{A}$ has no content, and $\mathcal{A} \subseteq \mathcal{U}$ follows from Theorem 1.37.

Remark 1.53. One can glue the tori $\operatorname{Spec} \mathbb{Z}\left[z_1^{\pm}, \ldots, z_n^{\pm}\right]$ together to produce $\mathcal{A}$ in some cases. This explains how Remark 1.50 generalizes our definition of a cluster variety.

In fact, in the acyclic case, these bounds collapse.

Theorem 1.54 (Berenstein–Fomin–Zelevinsky)**.** If $\mathcal{A}$ is an acyclic cluster algebra, then $\mathcal{L} = \mathcal{A} = \mathcal{U}$.

Sketch. One shows $\mathcal{U} \subseteq \mathcal{L}$ using the intersection lemma discussed in Theorem 1.37. ■

Example 1.55. From Example 1.26, one has that $\mathcal{A} = \mathbb{Z}[x_1, x_2, (x_2+1)/x_1, (x_1+1)/x_2]$. Indeed, the last cluster variable is $(x_1 + x_2 + 1)/(x_1x_2)$, which is $(x_2 + 1)/x_1 \cdot (x_1 + 1)/x_2 - 1$.

For example, one can now say something about friezes.

Definition 1.56. Fix an $n \times n$ generalized Cartan matrix. Then for each i, define the polynomial

$$f_{C,i} := x_i y_i - \prod_{j<o} x_j^{|c_{ji}|} - \prod_{j>i} x_j^{|c_{ji}|}$$

in $\mathbb{Z}[x_1, \ldots, x_n, y_1, \ldots, y_n]$. We now define $Y_1^{\text{frieze}}(C)$ to be the n-dimensional subvariety cut out by these functions. We may abbreviate $Y_1^{\text{frieze}}(C)$ to $Y_1^{\text{frieze}}(\Delta)$, where Δ denotes the generalized Dynkin type.

Remark 1.57. The notation Y_1 is intended to be similar to the notation of a modular curve.

Remark 1.58. If $\mathcal{A}$ is an acyclic cluster algebra, then it turns out that $\operatorname{Spec} \mathcal{A} \cong Y_1^{\text{frieze}}(C)$.

In fact, the remark upgrades as follows.

Theorem 1.59 (de Saint Germain–Huang–Lu)**.** Fix a generalized $n \times n$ generalized Cartan matrix C. Then there is a bijection between $Y_1^{\text{frieze}}(C)(\mathbb{N})$ and $\operatorname{Frieze}(C, \mathbb{N})$, which sends the variables $(x_1, \ldots, x_n)$ to the first column of the frieze. Here, $Y_1^{\text{frieze}}(C)(\mathbb{N})$ means that all the variables $(x_1, \ldots, x_n, y_1, \ldots, y_n)$ must get sent to positive integers.

Remark 1.60. The values of the other variables $y_1, \ldots, y_n$ can also be seen in other entries of the corresponding frieze

Thus, $Y_1^{\text{frieze}}(C)$ can be understood as a moduli space of friezes.

1.3.2 Diophantine Geometry

We will get some motivation from the following comparison of trichotomies.

- Fano varieties have $-K_X$ ample.
- Calabi–Yau varieties have K_X trivial.
- General type varieties have K_X big.

On the other hand, one has some results for curves.

- Fano cures have genus zero, which have a dense subset of points after a field extension.
- Calabi–Yau curves have genus one, which are elliptic curves (once attaining a point), which remain hard to understand.
- General type curves have genus at least two, which have finitely many points.

The first point produces the following conjecture.

Conjecture 1.61 (Campana)**.** A Fano variety X over a number field K admits a finite extension L for which $X(L)$ is Zariski dense.

The last point produces the following conjecture.

Conjecture 1.62 (Bombieri–Lang)**.** Fix a variety X of general type over a number field K. Then $X(K)$ is not Zariski dense.

Remark 1.63. There are similar statements for integral points.

The Calabi–Yau case remains hard to understand; for example, one can hunt for infinite orbits, but there are other ways to find rational points. For example, there is a conjecture that a K3 surface admitting local points and no Brauer–Manin obstruction admit rational points.

This trichotomy is relevant for us because our cluster varieties turn out to live in the interesting Calabi–Yau case.

Theorem 1.64 (Gross–Hacking–Keel)**.** Every cluster variety is log Calabi–Yau.

Idea. One can glue the tori together with the "boundary" cut out by $x_i = 0$. This gluing process allows one to glue together the holomorphic n-forms

$$\Omega_\Sigma = \frac{dx_1}{x_1} \wedge \cdots \wedge \frac{dx_n}{x_n}$$

on the cluster variety. Then Ω_Σ vanishes nowhere on the interior. The adjective "log" is present in the theorem because the resulting n-form Ω may have logarithmic poles on the boundary. ■

1.3.3 Counting Friezes

Note that it is not so hard to find rational points on $\operatorname{Spec} \mathcal{A}$, so we will focus on the harder question of finding integral points or even positive integral points, where a positive integral point is one in which the cluster variables go to positive integers.

It will help to have the following automorphism. The following is due to Kontsevich–Soibelman, Keller, Fock–Goncharov, Gross–Hacking–Keel, and Fomin–Zelevinsky.

Theorem 1.65. Fix a generalized Dynkin type Δ.

(a) There is a volume-preserving automorphism $\tau\colon Y_1^{\text{frieze}}(\Delta) \to Y_1^{\text{frieze}}(\Delta)$.

(b) Furthermore, τ is periodic if and only if Δ is finite; and if Δ is finite, then the period divides $h+2$, where h is the Coxeter number of Δ.

Remark 1.66. This τ is given by horizontally translating a frieze one column. From here, one still must check that τ descends to a regular automorphism on the cluster variety, preserves volumes, and is periodic. None of these checks are particularly easy.

Remark 1.67. This τ has many names: it can be the Donaldson–Thomas transformation, the maximal green sequence, the Auslander–Reiten translation, or the Fomin–Zelevinsky twist.

We can use this τ to understand positive integral points.

Theorem 1.68 (Zhang)**.** Fix a generalized Dynkin type Δ. Then $Y_1^{\text{frieze}}(\Delta)(\mathbb{N})$ is finite if and only if Δ is finite type, and the values are given by the table in Theorem 1.20.

Sketch. Let's begin with the case where Δ is infinite. Here are two arguments.

- Following Morier–Genoud, one can construct many maps $\mathcal{A} \to \mathbb{Z}$, basically by sending one collection of cluster variables to 1. There are infinitely many sets of cluster variables, so this produces infinitely many friezes.
- Alternatively, as soon as we have a single positive integral point (by sending cluster variables to 1), one can use Theorem 1.65 to show that the orbit of this point has infinite size. Roughly speaking, one knows that τ preserves positivity and integrality, and one can show that the size of the point grows.

We now turn to the finite type case. Define $Y_0^{\text{frieze}}(\Delta)$ to be the quotient of $Y_1^{\text{frieze}}(\Delta)$ by $\langle \tau \rangle$, which is a quotient by a finite group because τ now has finite order. We let $\mathcal{O}_\tau(x)$ denote the image of $x \in Y_1^{\text{frieze}}(\Delta)$.

Now, view $\log \mathcal{O}_\tau(x)$ as a vector and then apply the matrix C to it. It was shown by Muller that $C \cdot \log \mathcal{O}_\tau(x)$ is in a bounded set. However, Lusztig–Tits showed that C^{-1} has positive rational coefficients if and only if C is finite type, so one finds that the coordinates of $\log \mathcal{O}_\tau(x)$ are bounded by the coordinates of $C^{-1}(1, \ldots, 1)$. This shows that there can only be finitely many orbits $\mathcal{O}_\tau(x)$, so there are only finitely many points because τ has finite order!

It remains to actually count the number of friezes. For example, for $\Delta = E_8$, the bound provided by the previous paragraph is on the order of 10^{50} for the height of the orbit, which is much too large. We will do this with the following trick.

Lemma 1.69. Fix a finite Dynkin type Δ. Let i be a vertex in Δ of degree 1. Then the intersection of $Y_1^{\text{frieze}}(\Delta)$ with $x_i = 1$ is isomorphic to $Y_1^{\text{frieze}}(\Delta \setminus \{1\})$.

Proof. One can see this explicitly on the level of the equations. Alternatively, this amounts to saying that a frieze whose bottom row is all 1s is simply a frieze for the smaller Dynkin type. ■

Thus, we may assume that all nodes i of degree 1 have $x_i \geq 2$. From here, a long naïve calculation over a few days can complete the proof. For example, in E_8, there are only four new friezes not from a smaller Dynkin type. ■

Remark 1.70. It is an interesting question to find the number of τ-orbits among the friezes. Professor Zhang has threatened to put this on the homework.

Example 1.71. It turns out that $Y_1^{\text{frieze}}\left(\left[\begin{smallmatrix} 2 & -a \\ -b & 2 \end{smallmatrix}\right]\right)$ can be presented with the equation

$$xyz = (x^a + 1)^b + y.$$

Such equations appear in the Mordell–Schinzel conjecture, which dates back to a 1952 paper of Mordell. Using our developed theory, we can see that this equation has 5 points if $ab = 1$, 6 points if $ab = 2$, and 9 points of $ab = 9$, and there are infinitely many points if $ab \geq 4$. More generally, if $G(x, y)$ has positive integral coefficients, we are able to determine when $xyz = G(x, y)$ should have infinitely many positive integral solutions. For example, there are infinitely many when $\deg_x G$ and $\deg_y G$ are both at least three and has non-vanishing constant term.

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